

MATTHEW H. TAYLOR

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Research Interests

Data-driven control, reinforcement learning, dynamical systems, physics-informed machine learning

EDUCATION

Ph.D. in Computer Science and Engineering

University of California, San Diego, GPA: 4.0

Sept. 2025 – Est. June 2029

M.S. in Computer Science and Engineering

University of California, San Diego, GPA: 4.0

Sept. 2023 – June 2025

B.S. in Aerospace Engineering

University of Illinois at Urbana-Champaign, GPA: 3.93

Aug. 2019 – May 2023

RESEARCH AND WORK EXPERIENCE

UC San Diego

Graduate Researcher

Advisor: Sicun Gao

La Jolla, CA, USA

March 2024 –

- Improving learning-based design for high-dimensional control problems using reinforcement learning and generative modeling. Applications include balance and locomotion policies for the Unitree G1 humanoid.
- Fine-tuned the pi-0.5 vision-language-action model on a custom 3D-printed robot arm, collecting teleoperated demonstrations to efficiently automate pick-and-place of objects.
- Built an approximate neural Lyapunov controller for pendulum and cartpole Gymnasium environments, achieving a larger empirically validated region of attraction than an LQR baseline.

NASA Jet Propulsion Laboratory

Machine Learning Research Intern

Pasadena, CA, USA

June 2024 – Nov. 2024

- Leveraged self-supervised contrastive learning techniques to create an ocean synthetic-aperture radar foundation model and finetuned on ocean science object detection and semantic segmentation tasks.
- Developed a versatile, dataset and model-agnostic semantic segmentation and object detection training codebase with integration of Weights and Biases for visualizing and monitoring model performance.
- Curated a dataset with bounding box and polygon mask annotations for over 7,000 submesoscale ocean eddies in the western Mediterranean Sea in correspondence with oceanography experts.

Dolby Laboratories

Computer Vision/Machine Learning Engineer Intern

San Francisco, CA, USA

June 2023 – Aug. 2023

- Profiled video content using skin segmentation, noise, face, and text detection models to give suggestions for when to use precision detail, a local image adjustment algorithm created for Dolby Vision.
- Updated Dolby Vision's internal C++ to Python bridge by adding hooks for local enhancement parameters and produced a modular PyQt GUI to visualize content mapped Dolby Vision content and recommendations.

Maxar Technologies (now Vantor)

Research and Development Intern

Westminster, CO, USA

May 2022 – Aug. 2022

- Applied deep learning and semantic segmentation methods to detect building footprints from high-resolution and multispectral satellite imagery using PyTorch for training.
- Researched how the addition of the near infra-red band (NIR) improves segmentation performance.
- Developed model-agnostic visualizations in TensorBoard for the R&D Lab's custom training framework.

University of Illinois at Urbana-Champaign

Undergraduate Researcher

Advisor: Huy Tran

Champaign, IL, USA

May 2021 – May 2023

- Created a CNN to detect objects and transform a robot's camera view into a top-down grid world in simulation using Gazebo and ROS and deployed in the real-world using the NVIDIA JetBot AI kit.
- Explored useful applications for brain computer interfaces and human-agent teaming, applying machine learning algorithms to EEG data to improve a human-agent collaborative environment.
- Created an experiment with multiple participants and developed a multiclass SVM to classify LEDs flashing at different frequencies based on EEG signals using steady-state visual-evoked potentials.

TEACHING

University of California, San Diego

Computer Science and Engineering Department

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| • Teaching Assistant, AI Search and Reasoning (CSE 150B) | March 2025 – June 2025 |
| • Teaching Assistant, Search and Optimization (CSE 257) | Jan. 2025 – March 2025 |
| • Mentor TA, Teaching Methods in Computer Science (CSE 599) | Sept. 2024 – Dec. 2024 |
| • Teaching Assistant, Intro to Programming (CSE 8A) | Jan. 2024 – March 2024 |

AWARDS

- Outstanding Graduate/Professional Student Group award at UC San Diego (2026)
- Summa Cum Laude in Aerospace Engineering at University of Illinois at Urbana-Champaign (2023)
- James Scholar at University of Illinois at Urbana-Champaign (2020-2023)
- State Farm National Merit Scholar (2019)

PUBLICATIONS [[Google Scholar](#)]

*Equal Contribution

1. M.H. Taylor, Hongzhan Yu, Sicun Gao, Robustness Margins for Local Visual Perturbations in Vision-Language-Action Policies with Flow Matching, ICLR 2027 [*Under Review*]
2. Taylor, M.H., “Advantage Regularization for Imitating Complex Control Policies”, MS Thesis, University of California, San Diego, 2025.

RELEVANT COURSEWORK

University of California, San Diego

- Search and Optimization (CSE 257)
- Algorithms (CSE 202)
- Machine Learning for Physical Applications (ECE 228)
- Random Processes (ECE 250)
- Numerical Methods for Differential Equations (MAE 290B)
- Parallel Computing (CSE 260)
- Introduction to Robotics (CSE 276A)

University of Illinois at Urbana-Champaign

- Machine Learning (CS 446)
- Introduction to AI (CS 440)

- Probability and Statistics (CS 361)
- Numerical Methods (CS 357)
- Aerospace Control Systems (AE 353)
- Autonomous Systems Lab (AE 483)
- Aerospace Dynamical Systems (AE 352)

TECHNICAL SKILLS

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| <ul style="list-style-type: none"> • Python • C • C++ • Java • Git | <ul style="list-style-type: none"> • Unix • IsaacLab/mjlab/mujoco • PyTorch/TensorFlow • Distributed GPU Training |
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COMMUNITY SERVICE

Volunteer Dog Walker

Del Mar Dog Rescue

Del Mar, CA, USA
2024 –

- Volunteered to walk dogs at a local rescue on weekends.

Softball Coach

Special Opportunities Available in Recreation

Bloomington, IL, USA
2017 – 2019

- Coached children and adults with disabilities in a yearly softball league.

Buddy

The Miracle League of Central Illinois

Normal, IL, USA
2017 – 2019

- Helped children and adults with disabilities experience the joy of playing baseball.